Data Engineering

Azure Databricks Spark vs. SQL Functions

Cheat Sheet

Cloud with Garvit

1. Syntax Basics

Aspect	Spark (DataFrame API)	SQL (Spark SQL API)
Language Used	Python, Scala, Java	SQL
Execution Style	Functional (methods on DataFrame object)	Declarative (SQL queries)
Ease of Use	Ideal for programmatic logic	Familiarity for SQL users

2. Select and Filter Operations\

Operation	Spark Example (PySpark)	SQL Equivalent
Select Columns	df.select("column1", "column2")	SELECT column1, column2 FROM table
Filter Rows	df.filter(col("column") > 10)	SELECT * FROM table WHERE column > 10
Alias Columns	df.select(col("column").alias("alias"))	SELECT column AS alias FROM table

3. Aggregations

Operation	Spark Example (PySpark)	SQL Equivalent
Group By and Aggregate	df.groupBy("column").agg(avg("value"))	SELECT column, AVG(value) FROM table GROUP BY column
Count Rows	df.count()	SELECT COUNT(*) FROM table
Aggregate Functions	df.agg(sum("value"), max("value"))	SELECT SUM(value), MAX(value) FROM table

4. Joins

Operation	Spark Example (PySpark)	SQL Equivalent
Inner Join	df1.join(df2, "key", "inner")	SELECT * FROM df1 INNER JOIN df2 ON df1.key = df2.key
Left Join	df1.join(df2, "key", "left")	SELECT * FROM df1 LEFT JOIN df2 ON df1.key = df2.key
Cross Join	df1.crossJoin(df2)	SELECT * FROM df1 CROSS JOIN df2

5. Sorting

Operation	Spark Example (PySpark)	SQL Equivalent
Order Ascending	df.orderBy("column")	SELECT * FROM table ORDER BY column ASC
Order Descending	df.orderBy(desc("column"))	SELECT * FROM table ORDER BY column DESC

6. String Functions

Operation	Spark Example (PySpark)	SQL Equivalent
Substring	df.select(substring("column", 1, 3))	SELECT SUBSTRING(column, 1, 3) FROM table
String Contains	df.filter(col("column").contains("value"))	SELECT * FROM table WHERE column LIKE '%value%'
String Replace	df.select(regexp_replace("column", "x", "y"))	SELECT REPLACE(column, 'x', 'y') FROM table

7. Date and Time Functions

Operation	Spark Example (PySpark)	SQL Equivalent
Current Date	df.select(current_date())	SELECT CURRENT_DATE()
Add Days	df.select(date_add(col("date"), 5))	SELECT DATE_ADD(date, 5) FROM table
Date Difference	df.select(datediff(col("end"), col("start")))	SELECT DATEDIFF(end, start) FROM table

8. Window Functions

Operation	Spark Example (PySpark)	SQL Equivalent
Row Number	df.withColumn("row_num", row_number().over(Window.partitionBy("column")))	SELECT column, ROW_NUMBER() OVER (PARTITION BY column) AS row_num FROM table
Rank	df.withColumn("rank", rank().over(Window.partitionBy("column").orderBy("value")))	SELECT column, RANK() OVER (PARTITION BY column ORDER BY value) AS rank FROM table

9. Null Handling

Operation	Spark Example (PySpark)	SQL Equivalent
Drop Null Rows	df.na.drop()	DELETE FROM table WHERE column IS NULL
Replace Null Values	df.na.fill("value", ["column"])	SELECT COALESCE(column, 'value') FROM table

10. Miscellaneous

Operation	Spark Example (PySpark)	SQL Equivalent
Distinct Values	df.select("column").distinct()	SELECT DISTINCT column FROM table
Sample Rows	df.sample(fraction=0.1)	SELECT * FROM table TABLESAMPLE (10 PERCENT)
Create Temp View	df.createOrReplaceTempView("view_name")	Not applicable in SQL directly